Robert Curran

Staff AI/ML & Platform Engineer

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PROFESSIONAL SUMMARY

Staff AI/ML & Platform Engineer with 10+ years of experience building distributed services, data pipelines, and ML infrastructure. At Amazon, built **multimodal training-data pipelines for Nova** and **model-customization workflows using SageMaker HyperPod**, spanning data preparation, distributed training, fine-tuning, and evaluation. Hands-on with **Python, PyTorch, and distributed GPU systems**, with a focus on training throughput and pipeline reliability. Earlier work includes backend services and data processing for Amazon's search, inventory, and supply chain operations.

EXPERIENCE

Amazon · Staff Software Engineer — Seattle, WA | June 2023 – October 2026

▸ *AI/ML Engineer · Amazon Nova | Foundation Model Training & Multimodal AI* January 2025 – October 2026

Tech: *Python, Java, PyTorch, CUDA, NCCL, Transformer Architecture, Distributed Training, SFT, RLHF, PPO, Amazon Bedrock, Amazon S3*

- Built preprocessing and loading pipelines for **text, image, and video datasets**, increasing pipeline throughput by **40%** through parallel preprocessing and batch prefetching.
- Reduced training stalls by overlapping **data loading with GPU execution**, increasing training throughput by **32%**, measured in tokens per second on the same model and hardware configuration.
- Added dataset validation and filtering for **corrupted media, invalid records, and inconsistent metadata**, reducing data-related training failures by **35%**.
- Improved dataset preparation and evaluation for **SFT and RLHF experiments**, contributing to an **18% relative increase in instruction-following pass rate** against the previous production checkpoint.
- Owned the **multimodal training-data pipeline** from design through rollout, coordinating with research and infrastructure teams on data contracts, throughput targets, and failure monitoring.

▸ *AI Platform Engineer · SageMaker AI / HyperPod | Model Training & Customization (Nova Forge)* June 2023 – January 2025

Tech: *Python, PyTorch, Amazon SageMaker AI, SageMaker HyperPod, Amazon S3, Amazon EC2, NVIDIA GPUs, AWS Trainium, Distributed Training, Reinforcement Learning*

- Built customization workflows that loaded **intermediate Nova checkpoints** and configured datasets and training parameters for pre-training, mid-training, and post-training.
- Implemented dataset conversion, validation, and configurable mixing of **enterprise datasets with Nova-curated data** for continued pre-training and domain adaptation.
- Integrated **CPT and SFT recipes with SageMaker HyperPod**, standardizing distributed job configuration, dataset paths, and checkpoint output locations across customization runs.
- Integrated evaluation runs comparing customized checkpoints with their starting models on **domain-task accuracy and general instruction-following**, surfacing regressions before model selection.
- Owned the **training-recipe integration and rollout**, working with research and infrastructure teams to resolve configuration mismatches and document reproducible customization workflows.

Amazon · Software Engineer II / Senior Software Engineer — Seattle, WA | June 2018 – June 2023

Amazon Stores – Supply Chain Optimization Technologies (SCOT) | Multi-Echelon Inventory Planning

Tech: *Java, Python, Scala, Apache Spark, Kafka, AWS, S3, DynamoDB*

- Built **event-driven inventory synchronization services**, reducing the time for inventory updates to reach downstream fulfillment systems from **12 minutes to 90 seconds**.
- Reworked **Spark pipelines for inventory and demand data**, tuning partitioning and parallel execution to cut daily processing time from **4.5 hours to 1.2 hours**.
- Automated **inventory reconciliation and exception handling**, routing mismatched records for investigation and reducing manual corrections across fulfillment facilities.
- Added **deduplication and retry handling** to inventory consumers, preventing repeated events from applying the same inventory update more than once.
- Built **pipeline monitoring and recovery workflows** to detect delayed updates, identify failed processing stages, and replay affected data.
- Owned **service interfaces and production rollout** for inventory pipeline changes, coordinating with planning and fulfillment teams on data requirements and downstream dependencies.

- Implemented **data validation checks** for missing fields, invalid inventory quantities, and inconsistent product identifiers, isolating defective records before downstream processing.
- Built **pipeline monitoring and recovery workflows** to detect delayed updates, identify failed processing stages, and replay affected data.
- Developed **integration and load tests** for inventory services, covering duplicate events, downstream failures, and peak traffic before production releases.

Amazon · Backend Software Engineer — Seattle, WA | September 2016 – June 2018

Amazon Stores – Search & Discovery | Product Search & Ranking
Tech: *Java*, *Python*, *C++*, *AWS*, *DynamoDB*, S3, SQS, *Elasticsearch*

- Optimized **product-retrieval services** through query batching, caching, and parallel retrieval, reducing **p95 query latency by 35%**.
- Reworked **catalog indexing pipelines** to process incremental updates asynchronously, reducing **index refresh time by 72%**.
- Reduced redundant **DynamoDB reads** in search-ranking workflows by coalescing repeated requests and reusing cached results.
- Added **retry handling and failed-message reprocessing** to indexing consumers, allowing interrupted catalog updates to recover without rerunning the full pipeline.
- Built **load tests and service monitoring** for retrieval and indexing changes, validating latency and update freshness before production rollout.

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch, Transformer Architecture, Large Language Models (LLMs), Foundation Models, Multimodal AI, Model Fine-Tuning, Model Optimization
Model Training & Alignment: Pre-Training, Continued Pre-Training (CPT), Mid-Training, Supervised Fine-Tuning (SFT), Reinforcement Learning from Human Feedback (RLHF), PPO, GRPO, Reward Modeling, Model Alignment, Model Evaluation
Distributed Training & Computing: Distributed Training, CUDA, NCCL, GPU Computing, Data Parallelism, Model Parallelism, Training Optimization, Distributed Data Processing
AI & Cloud Infrastructure: Amazon SageMaker AI, SageMaker HyperPod, Amazon Bedrock, Amazon EC2, Amazon S3, AWS Trainium, NVIDIA GPUs
Data Engineering & Backend Systems: Apache Spark, Apache Kafka, DynamoDB, Elasticsearch, Event-Driven Architecture, Data Pipelines, Distributed Systems, High-Throughput Services

EDUCATION

State University of New York at Oswego
Bachelor of Science in Computer Science
August 2012 – May 2016